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**Regulation: R25C25****Course Code: 25PY101-7**

I B.Tech II Semester Regular Examinations – June, 2026

ENGINEERING PHYSICS

(CSE/ECE/ECE(VLSI))

Time: 150 Min**Max. Marks:60M****SECTION – A****Answer all Four questions****4X8M=32M**

1.

- a) How can Hall Effect determine whether a semiconductor is n-type or p-type? [1M]
- b) Apply the knowledge of Electric field and Lorentz force to obtain an expression for Hall voltage developed in an extrinsic semiconductor, when the magnetic field is applied perpendicular to the plane of the slab. [4M]
- c) A semiconductor carries a current of 5 mA placed transverse in a magnetic field of 0.2 T. The charge carrier concentration $n = 8 \times 10^{22} \text{ m}^{-3}$ and the thickness $t = 0.5 \text{ mm}$. Estimate the Hall voltage. [3M]

2.

- a) Elucidate the physical significance of the wavefunction ψ and it's probabilistic interpretation. [3M]
- b) Discuss the wave functions and probability density for particle in an infinite potential well for first three allowed states. [3M]
- c) An electron is bound in one dimensional potential well of width 0.18 nm. Find the energy value in eV of the ground state. [2M]

3.

- a) Describe the role of depletion region in the conversion of light energy into electric energy. [3M]
- b) Write the expressions for the Fill factor and Efficiency. [2M]
- c) Calculate the efficiency of a solar cell when the incident light energy is 1000 W/m^2 . The current flowing in the solar cell is when $R = 0$ is 20 mA and the voltage across its ends is 2.5 V when $R = \infty$. The Fill factor of the solar cell is given by 0.85 and the area is 0.04 m^2 . [3M]

4.

- a) Write the postulates of the Quantum free electron theory. [3M]
- b) What is Fermi-Dirac distribution function and explain its variation of the Fermi-Dirac distribution function at $T=0\text{K}$ for $E < E_F$ and $E > E_F$ and $T > 0\text{K}$ at $E=E_F$. [3M]
- c) Calculate the temperature at which there is 1% probability of finding the electron having energy 1eV above the Fermi level? [2M]

SECTION - B**Answer all Two questions****2X14M=28M**

5.

- a) Starting from the expressions of electron and hole concentrations of an intrinsic semiconductor and obtain the expressions for the Fermi level. [5M]
- b) The bandgap of the semiconductor is 1.1eV and $m_h^* = 0.28m_0$ and $m_e^* = 0.12 m_0$. Calculate the Fermi level at room temperature. [2M]
- c) Starting from the expressions of electron and hole concentrations of an intrinsic semiconductor obtain an expression for the electrical conductivity. [4M]
- d) An intrinsic semiconductor has a resistance of 300Ω at a room temperature, if its band gap energy is 0.7 eV calculate its resistance at 100K. [3M]

6.

- a) What is numerical aperture? Apply Snell's law to obtain an expression for numerical aperture in terms of refractive indices of core and cladding. [5M]
- b) The numerical aperture of an optical fiber is 0.2 when surrounded by air. Determine the refractive index of its core given the refractive index of cladding as 1.59. Also find the acceptance angle. [2M]
- c) Explain the necessity of stimulated emission for obtaining the LASER? [2M]
- d) What are semiconductor diode lasers? Describe with energy band diagram the working of semiconductor diode laser. Mention the uses of diode lasers. [5M]